Marmara Üniversitesi
Mühendislik Fakültesi
Elektrik-Elektronik Mühendisliği

Laboratory Manual

EE 3011
Electronics I

Experiment (1)
Introduction to LTspice
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Experiment (1)
Introduction to LTspice

Objectives

- To simulate simple passive circuits using DC and Transient analysis.
- To be able to generate LTspice results using plots for various circuit variables.

Pre-lab Work

Before starting this experiment, you need to study the document "Introduction to LTspice IV with Examples", by Nicholas Lombardo, Linear Technology, 2015.

Circuits

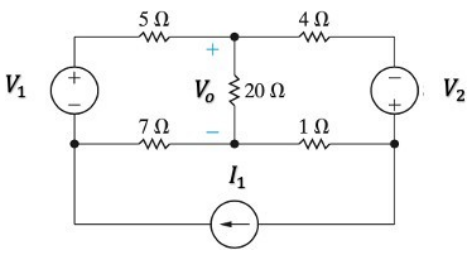


Figure 1: A DC circuit.

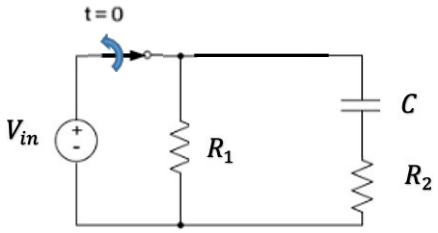


Figure 2: Transient circuit.

Procedure

Part (1): DC Analysis.

1. Connect the circuit in figure 1. Use the voltage and current sources values given by the lab instructor.
2. Using simulation, find the DC voltages and currents for all resistors. Record results in table 1.
3. Now make source V_1 as a variable voltage source and use the DC Sweep function to produce the transfer function curve V_o versus V_1.

Part (2): Transient Analysis.

1. Connect the circuit in figure 2. Use the circuit elements values given by the lab instructor. Set V_{in} as a periodic rectangular signal with frequency f and maximum amplitude V_{max}. The switch is initially closed.
2. Simulate the steady-state response of the circuit. Produce the steady-state voltage waveforms (versus time) across the capacitor C and the resistor R_1. Put these curves together with the V_{in} curve on the same plot.
3. Now, as the switch is opened at t=0, simulate the circuit using the Transient analysis function and generate the voltage waveform (versus time) across the capacitor C and the resistor R_2.

Put these curves together with the V_{in} curve on the same plot. Also observe the waveform of the current through R_{ℓ}. From the curves, find the time constant of the circuit.

4. Change the value of f and repeat step 3.

Results and Analysis

Part (1): DC Analysis.

Table 1. DC Simulations in Part (1).

$V_1 = 10V$ $V_2 = 5V$ $I_1 = 3A$	Simulation	Calculation	% Difference
Voltage across 1 Ω resitor	0.81 V	0.81 V	%0
Current through 1 Ω resitor	0.81 A	0.81 A	%0
Voltage across 4 Ω resitor	8.75 V	8.57 V	%0.0018
Current through 4 Ω resitor	2.19 A	2.19 A	%0
Voltage across 5 Ω resitor	11.69 V	11.88	%0.0019
Current through 5 Ω resitor	2.33 A	2.34 A	%0.0001
Voltage across 7 Ω resitor	4.67 V	4.64 V	%0.0003
Current through 7 Ω resitor	0.6625 A	0.6625 A	%0
Voltage across 20 Ω resitor	3.02 V	2.7625 V	%0.003
Current through 20 Ω resitor	0.1475 A	0.1475 A	%0

1. Comment on the difference between simulation and theoretical values in table 1.

%Difference is below %0.003 in all the voltages and currents. Simulation results and the calculation results that I calculated from node voltage method are so close to each other and we can say they are same and I think we can neglect this error in the real life implementations.

Part (2): Transient Analysis.

Table 2. Transient Simulations in Part (2).

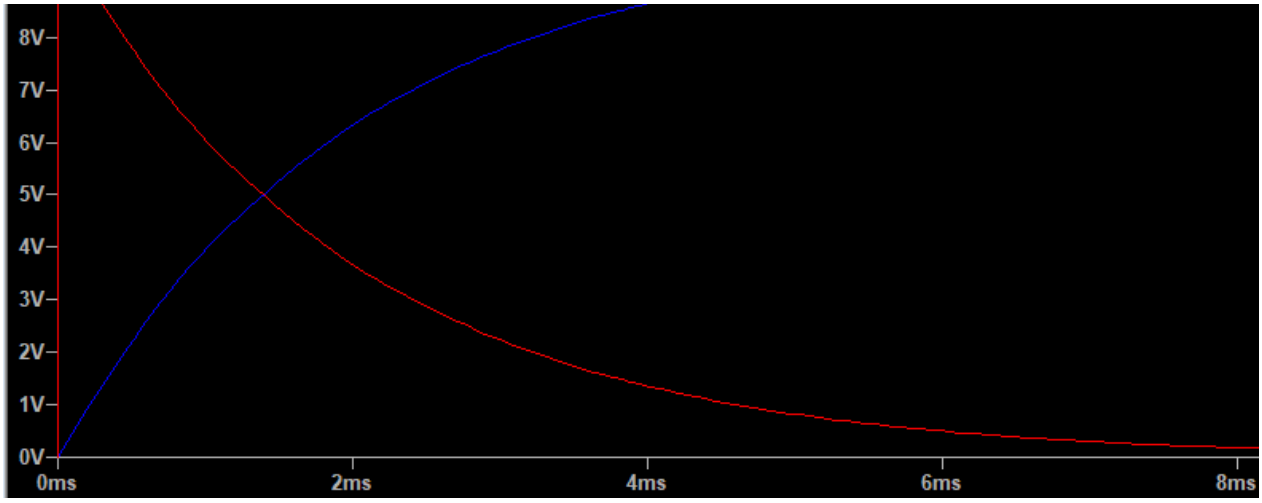
$R_1 = 1\text{k}\,\Omega \quad R_2 = 2\text{k}\,\Omega \quad C = 1\mu\text{F}$	Simulation	Calculation	% Difference
Circuit time constant	2 ms	0.66 ms	0.01

1. Comment on the difference between simulation and theoretical values in table 2.

   Difference is %0.01 and I think it is neglectable too.

2. State your observations on the simulated behavior of the given RC circuit, including the effect of changing the input frequency.

   Before t=0 there is no voltage and current change in the circuit because the capacitor is empty. Right after the switch is opened, the voltage graph of Vin(green), C(blue) and R2(red) is:



   Capacitor loads itself and voltage of the resistor connected to the capacitor decreases exponentially. Time constant is around 2 ms. When I changed the frequency adjusting the Tperiod and Ton values, I get the same microsecond value. The formula is R*C and we expected the same result.